SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION **COURSE CODE:** DSA0204
COURSE OUTCOME COVERED

NAME: VM SAI BHARGAV **REG NO: 192324126**

***CO4:** Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

Answer:

(a) Precision and Recall

Precision = TP / (TP + FP) Recall = TP / (TP + FN)

Model A (Viola-Jones): Precision = 80/(80+20) = 80/100 = 0.80 (80%) Recall = 80/(80+30) = 80/110 = 0.727 (72.7%)

Model B (YOLO): Precision = 90/(90+15) = 90/105 = 0.857 (85.7%) Recall = 90/(90+20) = 90/110 = 0.818 (81.8%)

(b) Comparison and Conclusion

Model B (YOLO) outperforms Model A on both precision (85.7% vs 80%) and recall (81.8% vs 72.7%). YOLO produces fewer false positives and fewer false negatives relative to its total predictions, so it is the better model for object detection in this comparison.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

Answer:

(a) Average Error

Method A: (2+3+2)/3 = 7/3 = 2.33 pixels

Method B: (3+5+4)/3 = 12/3 = 4.00 pixels

(b) Comparison and Conclusion

Method A has a lower average displacement error (2.33 px) compared to Method B (4.00 px) across all three frames. Method A is therefore more accurate for motion estimation in this dataset.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

Answer:

(a) Average Motion Magnitude

Method A: (5+6+5)/3 = 16/3 = 5.33 px/frame

Method B: (8+10+9)/3 = 27/3 = 9.00 px/frame

(b) Comparison and Conclusion

Method B consistently records higher magnitudes across all regions and captures stronger, more pronounced motion than Method A. However, higher magnitude alone does not confirm greater accuracy — if ground truth motion is closer to Method A's range, B may be over-estimating; assuming both methods are otherwise validated, Method B is more sensitive to and better at capturing strong motion.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters). Actual depths: A=2, B=3, C=4

Object	Stereo Vision	Structure from Motion
A	2	2.5
B	3	3.8
C	4	5

Answer:

(a) Absolute Error

Stereo Vision: |2.0-2|=0, |3.0-3|=0, |4.0-4|=0 → Average absolute error = 0.0 m

Structure from Motion: |2.5-2|=0.5, |3.8-3|=0.8, |5.0-4|=1.0 → Average absolute error = (0.5+0.8+1.0)/3 = 2.3/3 = 0.767 m

(b) Comparison and Conclusion

Stereo Vision matches the actual depth exactly for all three objects (zero error), while Structure from Motion shows increasing error as depth increases. Stereo Vision is clearly the more accurate method here, though it requires dual-camera hardware, whereas SfM's error grows with distance but works with a single camera.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

Answer:

(a) Accuracy

Traditional: 85/100 = 0.85 = 85%

Deep Learning: 92/100 = 0.92 = 92%

(b) Comparison and Conclusion

The Deep Learning system achieves 7 percentage points higher accuracy than the Traditional system (92% vs 85%), making it the better-performing pattern recognition system on this sample set.

6. Precision-Recall Trade-off Analysis

Model A: TP=70, FP=30, FN=20 Model B: TP=85, FP=25, FN=25

Answer:

(a) Precision and Recall

Model A: Precision = 70/(70+30) = 0.70 (70%) Recall = 70/(70+20) = 70/90 = 0.778 (77.8%)

Model B: Precision = 85/(85+25) = 85/110 = 0.773 (77.3%) Recall = 85/(85+25) = 85/110 = 0.773 (77.3%)

(b) Comparison and Conclusion

Model A shows a gap of about 7.8 percentage points between precision and recall, while Model B's precision and recall are almost identical (77.3% each). Model B is therefore the more balanced model, offering an even trade-off between false positives and false negatives, which is desirable when neither error type should dominate.

7. F1-Score Comparison

Model A: Precision=0.8, Recall=0.6   Model B: Precision=0.75, Recall=0.75

Answer:

(a) F1-Score

F1 = 2 × (Precision × Recall) / (Precision + Recall)

Model A: F1 = 2×(0.8×0.6)/(0.8+0.6) = 0.96/1.4 = 0.686 (68.6%)

Model B: F1 = 2×(0.75×0.75)/(0.75+0.75) = 1.125/1.5 = 0.75 (75%)

(b) Comparison and Conclusion

Model B has a higher F1-score (0.75) than Model A (0.686), showing a better balance between precision and recall overall. Model B is the better-performing model when both false positives and false negatives matter equally.

8. Optical Flow Consistency

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

Answer:

(a) Average Motion

Method A: (4+5+6)/3 = 15/3 = 5.00

Method B: (6+8+9)/3 = 23/3 = 7.67

(b) Consistency and Stability

Variance A = $[(4-5)^2+(5-5)^2+(6-5)^2]/3 = (1+0+1)/3 = 0.667$

Variance B = $[(6-7.67)^2+(8-7.67)^2+(9-7.67)^2]/3 = (2.79+0.11+1.77)/3 = 1.556$

Method A has a lower variance (0.667) than Method B (1.556), indicating more consistent, stable frame-to-frame motion values, even though Method B records higher raw motion magnitude. Method A is therefore more stable.

9. Detection Accuracy Comparison

Results:

Model	Correct	Total
A	180	200
B	170	200

Answer:

(a) Accuracy

Model A: 180/200 = 0.90 = 90%

Model B: 170/200 = 0.85 = 85%

(b) Comparison and Conclusion

Model A achieves higher detection accuracy (90%) than Model B (85%) and is therefore the better-performing model in this comparison.

10. Error Rate Comparison

Results:

Method	Errors	Total
A	20	100
B	10	100

Answer:

(a) Error Rate

Method A: 20/100 = 0.20 = 20%

Method B: 10/100 = 0.10 = 10%

(b) Comparison and Conclusion

Method B has half the error rate of Method A (10% vs 20%) and is therefore the more reliable, better-performing method.

11. Motion Estimation Variance

Errors:

Frame	A	B
1	2	6
2	3	7
3	2	5

Answer:

(a) Mean Error

Method A: (2+3+2)/3 = 7/3 = 2.33

Method B: (6+7+5)/3 = 18/3 = 6.00

(b) Variability and Stability

Variance A = $[(2-2.33)^2+(3-2.33)^2+(2-2.33)^2]/3 = (0.11+0.44+0.11)/3 = 0.222$

Variance B = $[(6-6)^2+(7-6)^2+(5-6)^2]/3 = (0+1+1)/3 = 0.667$

Method A has both a lower mean error (2.33) and lower variance (0.222) than Method B (mean 6.00, variance 0.667), making Method A more accurate and more stable overall.

12. IOU Comparison

IOU values:

Object	A	B
1	0.6	0.8
2	0.7	0.85
3	0.65	0.9

Answer:

(a) Average IOU

Method A: (0.60+0.70+0.65)/3 = 1.95/3 = 0.65

Method B: (0.80+0.85+0.90)/3 = 2.55/3 = 0.85

(b) Comparison and Conclusion

Method B achieves a substantially higher average IOU (0.85) than Method A (0.65), indicating tighter, more accurate bounding-box overlap with ground truth. Method B provides better detection quality.

13. Recall Comparison in Detection

Results:

Model	TP	FN
A	60	40
B	80	20

Answer:

(a) Recall

Model A: 60/(60+40) = 60/100 = 0.60 = 60%

Model B: 80/(80+20) = 80/100 = 0.80 = 80%

(b) Comparison and Conclusion

Model B has a much higher recall (80%) than Model A (60%), meaning it misses fewer true positives. Model B is more sensitive and preferable when minimizing missed detections is important.

14. Depth Estimation Error (Squared)

Predicted vs Actual depth:

Object	A	B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

Answer:

(a) Squared Error

Object 1: SE_A = $(2.5-2)^2 = 0.25$ SE_B = $(2.2-2)^2 = 0.04$

Object 2: SE_A = $(3.5-3)^2 = 0.25$ SE_B = $(3.1-3)^2 = 0.01$

Object 3: SE_A = $(4.5-4)^2 = 0.25$ SE_B = $(4.2-4)^2 = 0.04$

Mean Squared Error A = (0.25+0.25+0.25)/3 = 0.25

Mean Squared Error B = (0.04+0.01+0.04)/3 = 0.03

(b) Comparison and Conclusion

Method B has a much lower Mean Squared Error (0.03) than Method A (0.25), showing that its depth predictions are consistently closer to the true values. Method B is significantly more accurate.

15. Recognition Precision Comparison

Results:

System	TP	FP
A	90	10
B	85	5

Answer:

(a) Precision

System A: 90/(90+10) = 90/100 = 0.90 = 90%

System B: 85/(85+5) = 85/90 = 0.944 = 94.4%

(b) Comparison and Conclusion

System B has a higher precision (94.4%) than System A (90%), producing proportionally fewer false positives among its predictions. System B performs better on this measure.

Code of Conduct:

I V M SAI BHARGAV (192324126) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V M SAI BHARGAV